$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & -\frac{Y_1 k^2}{4} & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & Y_2 k^2
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{Y_3 k^2}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{Y_3 k^2}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{k^2(\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_{16})}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(2^-)}
\end{array}
$$

Abbreviations used in matrices

$$Y_1 == 6\,\overset{(4)}{\kappa}_{1} + 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \;\&\&\; Y_2 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \;\&\&\; Y_3 == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \;\&\&$$

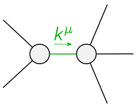
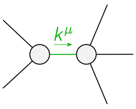
$$\mathrm{Det}(0^+) == -\frac{1}{4} k^2 (6\,\overset{(4)}{\kappa}_{1} + 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}) \;\&\&\; \mathrm{Det}(2^+) == \frac{1}{2} k^2 (2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16})$$

Lagrangian

$$
\begin{aligned}
&\overset{(4)}{\kappa}_{1}\, \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{15}\, \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \frac{1}{2}\overset{(4)}{\kappa}_{16}\, \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \\
&\overset{(4)}{\kappa}_{15}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} - \frac{1}{2}\overset{(4)}{\kappa}_{16}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} + 2\,\overset{(4)}{\kappa}_{16}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} - \\
&\overset{(4)}{\kappa}_{15}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\overset{(4)}{\kappa}_{16}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - 2\,\overset{(4)}{\kappa}_{15}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\, \partial_\delta \mathcal{K}_{\chi\beta}{}^{\delta} - \\
&\overset{(4)}{\kappa}_{16}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\, \partial_\delta \mathcal{K}_{\chi\beta}{}^{\delta} - \overset{(4)}{\kappa}_{15}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\, \partial_\delta \mathcal{K}_{\alpha\beta\chi}\, \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16}\, \partial_\delta \mathcal{K}_{\beta\alpha\chi}\, \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#2\,\alpha\beta}_{1^+} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\delta\partial_\chi\partial^\beta \mathcal{J}^{\chi\alpha\delta}$
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial_\chi\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\alpha\beta\chi}$
Total # constraint(s):	12	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$
	2	0	$-\frac{1}{2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$

Resolved unitarity condition(s):	(Demonstrably impossible)